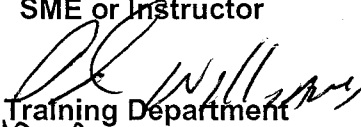
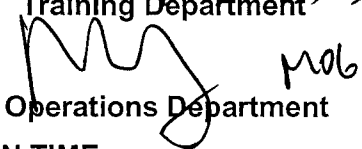


OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	SALEM		
SYSTEM:	Component Cooling System		
TASK:	Take Corrective Action For A Component Cooling Water System Abnormality		
TASK NUMBER:	N1140080401		
JPM NUMBER:	14-01 NRC Sim g		
ALTERNATE PATH:	☒	K/A NUMBER:	008 A4.01
APPLICABILITY:		IMPORTANCE FACTOR:	3.3 3.1
EO ☐	RO ☒	STA ☐	SRO ☒
EVALUATION SETTING/METHOD:	Simulator - Perform		
REFERENCES:	S2.OP-AB.CC-0001, Rev. 14 Component Cooling Abnormality S2.OP-AB.RCP-0001, Rev. 21 Reactor Coolant Pump Abnormality S2.OP-AR.ZZ-0004, Rev. 27, Overhead Window D (all rev checked 9-23-15) S2.OP-AR.ZZ-0011, Rev. 60, Control Console 2CC1		
TOOLS AND EQUIPMENT:	None		
VALIDATED JPM COMPLETION TIME:	6 minutes		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	N/A		
Developed By:	G Gauding Instructor	Date:	8-20-15
Validated By:	S Bickhart / J Militti SME or Instructor	Date:	9-23-15
Approved By:	 Training Department	Date:	10-7-15
Approved By:	 Operations Department	Date:	10-9-15
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE:	SAT	UNSAT	
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:	DATE:		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Component Cooling System

TASK: Take Corrective Action For A Component Cooling Water System Abnormality

TASK NUMBER: N1140080401

SIMULATOR SETUP IC-247 100% power EOL. ***Ensure VCT level is sufficient such that a swap to RWST will not occur.

MALF: CC0361 23 CCW pump fails to start on low pressure

RT-1

MALF: CC0172B 22 Component Cooling Water Pump Trip

MALF: CC0172A 21 Component Cooling Water Pump Trip

Delay: 1 minute 30 seconds

Override: A604 OVDI 23 CCW pump start PB - OFF

INITIAL CONDITIONS: 100% power, EOL.

INITIATING CUE: You are the Reactor Operator. Respond to all alarms and indications.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:

1. Trip the Reactor.
2. Trip 21-24 RCPs.
3. Isolate Letdown and swap Charging Pump suction to the RWST.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Component Cooling System

TASK: Take Corrective Action For A Component Cooling Water System Abnormality

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		Note for Simulator Operator: The remaining CCW pump <u>must</u> be tripped PRIOR to the determination that the Rx must be tripped and RCPs stopped in either the Reactor Coolant Pump or Component Cooling System Abnormal Procedures. IF required, THEN remove the delay from 22 CCW pump trip prior to the operator tripping the reactor.	Note to Evaluator: The end result of this JPM will be the same whether the Component Cooling or Reactor Coolant Pump Abnormal Procedure is used. The Attachment in each procedure for Stopping RCPs is the same. S2.OP-AB.RCP-0001 steps start on page 9 S2.OP-AB.CC-0001 steps start on page 14		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Component Cooling System

TASK: Take Corrective Action For A Component Cooling Water System Abnormality

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		Simulator Operator: Insert RT-1 after operator assumes the watch. MALF: CC0172B 22 Component Cooling Water Pump Trip MALF: CC0172A 21 Component Cooling Water Pump Trip DELAY: 1 minute 30 seconds Note: IF operator announces intention to trip the Rx before the remaining CCW pump has tripped, then remove remaining delay time from the 21 CCW pump trip.	Reports 22 CCW pump has tripped. The following alarms will annunciate: • OHAs D20-D23 21/22/23/24 RCP BRG CLG WTR FLO LO. • RCP Thermal Barrier Return Lo Flow. • 21 and 22 CCW Header Lo Pressure Console alarms. Note: Each of the Alarm Response actions is contained here, as well as the actions contained in both S2.OP-AB.RCP-0001, Reactor Coolant Pump Abnormality, and S2.OP-AB.CC-0001, Component Cooling Abnormality. Either of the Abnormal Procedures may be entered directly, or through the Alarm Response Procedure to get to ABs.		
			Determines that the standby CCW pump has failed to auto start, and attempts to manually start it here or when directed by Abnormal Procedure. When attempted, reports standby CCW pump will not start.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Component Cooling System

TASK: Take Corrective Action For A Component Cooling Water System Abnormality

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	ARP D20-23		Refers to S2.OP-AR.ZZ-0004, Overhead Window D.		
	ARP D20-23 2.0		Determines there are no automatic actions associated with these alarms.		
	ARP D20-23 3.2	CHECK OPEN the following Component Cooling Water valves: A. 2CC117, RCP CC MOT OP INLET V. B. 2CC118, RCP CC MOT OP INLET VALVE C. 2CC136, RCP BRG MOT OP OUTLET VALVE D. 2CC187, RCP MOTOR CC MOTOR OPERATED OUTLET VALVE.	Determines the following Component Cooling Water valves are open on CC1: A. 2CC117, RCP CC MOT OP INLET V. B. 2CC118, RCP CC MOT OP INLET VALVE C. 2CC136, RCP BRG MOT OP OUTLET VALVE D. 2CC187, RCP MOTOR CC MOTOR OPERATED OUTLET VALVE.		
	ARP D20-23 3.3	MONITOR 21(22,23,24) RCP Motor Bearing temperatures. (T0413A, T0414A T0415A, and T0416A)	Monitors 21(22,23,24) RCP Motor Bearing temperatures. (T0413A, T0414A T0415A, and T0416A) using P-250 computer on the Reactor Coolant Pumps screen, which has 21-24 RCP bearing temperatures. Note: OHA D20-23 are identical except for bearing temperature instrument numbers.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Component Cooling System

TASK: Take Corrective Action For A Component Cooling Water System Abnormality

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	ARP D20-23 3.4	<u>IF</u> either of the following conditions exist, GO TO S2.OP-AB.RCP-0001(Q) Reactor Coolant Pump Abnormality: • 21(22,23,24) RCP Motor Bearing Temperature reaches 175°F, or • CCW flow can <u>NOT</u> be established within 5 minutes <u>AND</u> 21 RCP Motor Bearing temperatures are trending up.	Determines 21(22,23,24) RCP Motor Bearing Temperature has not reached 175° (will reach maximum of ~ 157 °F) Determines that RCP Motor Bearing temperatures are not trending up. (The temps will rise from ~141 to 157 quickly, within 3 minutes, and the curve of the line flattens out quickly. There IS still CCW flow going to the RCPs, and it WILL keep them from reaching required pump trip temperatures. <u>IF</u> pace of JPM performance results in it being determined that temperatures are trending up, <u>THEN</u> will go to S2.OP-AB.RCP-001 here.		
	ARP D20-23 3.5	MONITOR CC Surge Tank and Containment Sump Pump runs for leakage indication.	Determines CC Surge Tank and Containment Sump Pump runs do not indicate leakage.		
	ARP D20-23 3.6	<u>IF</u> CC Console alarms header pressure low, <u>OR</u> Surge Tank Level High Low, <u>THEN</u> INITIATE S2.OP-AB.CC-0001, Component Cooling Abnormality.	Initiates S2.OP-AB.CC-0001, Component Cooling Abnormality based on CC header Pressure low console alarms.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Component Cooling System

TASK: Take Corrective Action For A Component Cooling Water System Abnormality

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	ARP CC1 2.0	21 (22) CC HDR PRESSURE LO <u>Automatic Actions</u> – Starts the standby CC Pump	Determines that the standby CCW pump has failed to auto start,		
	ARP CC1 3.1	ENSURE the CC Pump is RUNNING or START standby CC Pump.	Attempts to manually start the standby CCW pump, and recognizes it will not start.		
	ARP CC1 3.2	<u>IF</u> the low pressure alarm remains, <u>THEN</u> GO TO S2.OP-AB.CC-0001(Q), Component Cooling system Abnormality.			
	AB RCP 3.1	INITIATE Attachment 1, Continuous Action Summary.	Initiates Attachment 1, Continuous Action Summary. Note: The first 2 bulleted CAS actions for 1) losing CCW flow to ALL RCPs 2) not restoring CCW flow within 5 minutes of initial loss are based on a loss of ALL CCW flow. However, if a conservative decision is made that the actions of Attachment 2, Stopping Reactor Coolant Pumps are applicable, then CUE: The CRS directs you to dispatch an operator to 23 CCW pump breaker first to see if there is any apparent reason why 23 CCW pump breaker did not close.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Component Cooling System

TASK: Take Corrective Action For A Component Cooling Water System Abnormality

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	AB RCP 3.2	MONITOR AND COMPARE RCP parameters in Attachment 3 to determine affected RCP(s) until malfunction is corrected or as directed by SM/CRS.	Monitors and compares RCP parameters in Attachment 3 on the Plant Computer, Control Console, and RP Panels, and determines that all RCPs are affected.		
	AB RCP 3.3	<u>IF</u> Component Cooling Water supply is causing the RCP(s) abnormal condition, <u>THEN</u>: A. ENSURE the following automatic CC valves supplying RCP(s) are OPEN. • 2CC117 <u>AND</u> 2CC118, RCP COOLING INLET • 2CC131 <u>AND</u> 2CC190, RCP THERMAL BARRIER VALVE • 2CC136 <u>AND</u> 2CC187, RCP BEARING OUTLET B. ENSURE CCW Heat exchanger(s) in service IAW S2.OP-SO.CC-0002(Q), 21 & 22 Component Cooling Heat Exchanger Operation.	Determines 2CC117 <u>AND</u> 2CC118, RCP COOLING INLET, 2CC131 <u>AND</u> 2CC190, RCP THERMAL BARRIER VALVE, and 2CC136 <u>AND</u> 2CC187, RCP BEARING OUTLET valves are all open on control console. Determines that CCW system operation prior to the event was normal, and CCW HX(s) are in service. Cue if required: <u>IF</u> verification of CCW HX operation is begun using S2.OP-SO.CC-0002, <u>THEN</u> cue: "21 & 22 CCW HX's are in service."		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Component Cooling System

TASK: Take Corrective Action For A Component Cooling Water System Abnormality

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		C. ENSURE RCP Seal Injection flows are 6-12 gpm per pump, <u>NOT</u> to exceed 40 gpm total RCP Seal Injection Flow.	Checks RCP Seal Injection flows are 6-12 gpm per pump, total RCP Seal Injection Flow adds up to $\leq$ 40 gpm.		
	AB RCP 3.4	<u>IF</u> any RCP Motor winding temperature, indicated on 2RP4 recorder, is $>235^{\circ}\text{F}$ for Hot Loop Conditions (Modes 1-4), <u>OR</u> $>248^{\circ}\text{F}$ for Cold Loop Conditions (Mode 5) <u>THEN</u> :	Determines all RCP Motor winding temperatures, indicated on 2RP4 recorder, are $<235^{\circ}$ for Hot Loop Conditions (Modes 1-4).		
	AB RCP 3.5	<u>IF</u> Loss of Seal Injection is causing RCP(s) abnormal condition, <u>AND</u> RCP Seal Leakoff is <2.5 gpm on any RCP, <u>THEN</u> :	Determines Loss of Seal Injection flow is not causing RCS(s) abnormal condition.		
	AB RCP 3.6	<u>IF</u> affected RCP Seal Water Leakoff is <0.8 gpm or is slowly approaching 6 gpm, <u>AND</u> affected Pump bearing/seal inlet temperatures are stable or lowering, <u>THEN</u> :	Determines affected RCP Seal Water Leakoff is >0.8 gpm and is not slowly approaching 6 gpm from console indications.		
			Note: 21 CCW pump will trip 1 minute and 30 seconds after 22 CCW pump trips. Determines last remaining Component Cooling water has tripped.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Component Cooling System

TASK: Take Corrective Action For A Component Cooling Water System Abnormality

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	AB RCP CAS 1.0	<u>IF AT ANY TIME</u> , any of the <u>validated</u> RCP conditions exist, <u>THEN GO TO</u> Attachment 2, Stopping Reactor Coolant Pumps:	Determines Component Cooling Water flow has been lost to all RCPs and goes to Attachment 2, Stopping Reactor Coolant pumps.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Component Cooling System

TASK: Take Corrective Action For A Component Cooling Water System Abnormality

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
* * * * *	AB.RCP Att. 2 1.0	<u>IF</u> Reactor Trip Breakers are CLOSED, <u>THEN</u>: 1.1 TRIP the Reactor. 1.2 STOP affected RCP(s). 1.3 <u>IF</u> RCP shutdown was due to RCP seal Leakoff flow > 6gpm, <u>THEN</u> simultaneously PERFORM the following: 1.4 <u>IF</u> RCP shutdown was due to a TOTAL LOSS of Component Cooling Water (including loss of CCW to the CVCS Letdown Heat Exchanger), <u>THEN</u> simultaneously PERFORM the following: • ISOLATE RCS letdown <u>AND</u> SWAP charging pump suction to the RWST by performing the following: a) CLOSE 2CV2 <u>AND</u> 2CV277 to isolate letdown. b) OPEN 2SJ1 <u>AND</u> 2SJ2 to swap charging pump suction to the RWST. c) CLOSE 2CV40 <u>AND</u> 2CV41 to isolate the VCT. • GO TO 2-EOP-TRIP-1, Reactor Trip or Safety Injection <u>AND</u> RETURN to this attachment, at the SM/CRS discretion.	Trips the Reactor using either trip handle. Depresses stop PBs for 21-24 RCPs. Determines RCP shutdown was not due to RCP seal Leakoff flow > 6gpm. Cue if required: IF operator continues with TRIP-1 Immediate Actions, THEN CUE to return to AB.RCP-1 and other operators will continue performing TRIP-1 after IA's are complete. Determines RCP shutdown was due to a TOTAL LOSS of Component Cooling Water and simultaneously performs the following: Depresses the close PB for 2CV2 <u>AND</u> 2CV277 to isolate letdown. Depresses the open PB for 2SJ1 <u>AND</u> 2SJ2 to swap charging pump suction to the RWST. Depresses the close PB for 2CV40 <u>AND</u> 2CV41 to isolate the VCT. Goes to 2-EOP-TRIP-1, Reactor Trip or Safety Injection. Terminate JPM when TRIP-1 immediate actions begin to be performed.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Component Cooling System

TASK: Take Corrective Action For A Component Cooling Water System Abnormality

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	AB.CC 3.1	INITIATE Attachment 1, Continuous Action Summary.	Initiates Attachment 1, Continuous Action Summary.		
	AB.CC 3.2	Is CCW Surge Tank level >58% and rising?	Determines CCW Surge tank level is < 58% and stable from console indication, goes to Step 3.13.		
	AB.CC 3.13	Is CCW Surge Tank level <42% and dropping?	Determines CCW Surge tank level is >42% and stable from console indication, goes to Step 3.40.		
	AB.CC 3.40	Is either of the following annunciators in alarm? • 21 CC HDR PRESSURE LO • 22 CC HDR PRESSURE LO	Determines both 21 and 22 CC HDR PRESSURE LO annunciators are in alarm.		
	AB.CC 3.41	START available CC Pumps as necessary to clear CC HDR PRESSURE LO alarms(s).	Determines no CCW pumps are available to be started to clear CC HDR PRESSURE LO alarms(s).		
			Note: 21 CCW pump will trip 1 minute and 30 seconds after 22 CCW pump trips. Determines last remaining Component Cooling water has tripped.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Component Cooling System

TASK: Take Corrective Action For A Component Cooling Water System Abnormality

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	AB.CC CAS 1.0	<u>IF AT ANY TIME</u> , any of the <u>validated</u> conditions exist, <u>THEN STOP</u> RCP's IAW Attachment 2, Stopping Reactor Coolant Pumps:	Determines Component Cooling Water flow has been lost to all RCPs and goes to Attachment 2, Stopping Reactor Coolant pumps.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Component Cooling System

TASK: Take Corrective Action For A Component Cooling Water System Abnormality

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
* * * * *		<u>IF</u> Reactor Trip Breakers are CLOSED, <u>THEN</u>: A. TRIP the Reactor. B. STOP affected RCP(s). C. Simultaneously PERFORM the following: <u>IF</u> a total loss of Component Cooling Water was the initiating event (including loss of CCW to the CVCS Letdown Heat Exchanger), <u>THEN ISOLATE</u> RCS letdown <u>AND SWAP</u> Charging Pump suction to the RWST as follows: a. CLOSE 2CV2 <u>AND</u> 2CV277 to isolate letdown. b. OPEN 2SJ1 <u>AND</u> 2SJ2 to swap charging pump suction to the RWST. c. CLOSE 2CV40 <u>AND</u> 2CV41 to isolate the VCT. <u>IF</u> a Component Cooling Water LEAK in Containment with 2CC113 <u>AND</u> 2CC215 closed was the initiating event, <u>THEN CLOSE</u> the following valves: <u>GO TO</u> 2-EOP-TRIP-1, Reactor Trip or Safety Injection.	Trips the Reactor using either trip handle. Depresses stop PBs for 21-24 RCPs. Cue if required: IF operator continues with TRIP-1 Immediate Actions, <u>THEN CUE to return to AB.RCP-1 and other operators will continue performing TRIP-1 after IA's are complete.</u> Depresses the close PB for 2CV2 <u>AND</u> 2CV277 to isolate letdown. Depresses the open PB for 2SJ1 <u>AND</u> 2SJ2 to swap charging pump suction to the RWST. Depresses the close PB for 2CV40 <u>AND</u> 2CV41 to isolate the VCT. Determines CCW leak in containment was not the initiating event. Goes to 2-EOP-TRIP-1, Reactor Trip or Safety Injection. Terminate JPM when TRIP-1 immediate actions begin to be performed.		

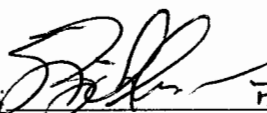
JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

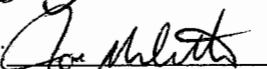
- Q S 1. Task description and number, JPM description and number are identified.
- Q S 2. Knowledge and Abilities (K/A) references are included.
- Q S 3. Performance location specified. (in-plant, control room, or simulator)
- Q S 4. Initial setup conditions are identified.
- Q S 5. Initiating and terminating Cues are properly identified.
- Q S 6. Task standards identified and verified by SME review.
- Q S 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- Q S 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 14 Date 10/6/10 AB.CC-6001
- Q S 9. Pilot test the JPM:

<u>21</u>	<u>7/3/8</u>	<u>AB.RCP-0001</u>
<u>27</u>	<u>4/24/15</u>	<u>AR.ZZ-6004</u>
<u>60</u>	<u>4/30/15</u>	<u>AR.ZZ-6011</u>

 - a. verify Cues both verbal and visual are free of conflict, and
 - b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor:  BICKHART

Date: 9/23/15

SME/Instructor:  M.J.H.

Date: 9-23-15

SME/Instructor: _____

Date: _____

INITIAL CONDITIONS:

100% power, EOL.

INITIATING CUE:

You are the Reactor Operator. Respond to all alarms and indications.